

Data overview

1. Sensor Reading Table — Column Overview

sensor table shape: (2526117, 5)

#	Column	Dtype	Null-count	Null %	Unique value	Range	percentile
1	timestamp	object	0	0	410165	-	-
2	machine_id	int64	0	0	8	1 - 8	2 : 5 : 8
3	sensor_id	object	0	0	48	-	-
4	value	float64	756370	29.9	17774	(-14) - 1036.16	9.56 : 171.76 : 209.06
5	unit	object	0	0	5	-	-

Range of timestamp column:
Start: 2024-01-01 06:09:00
End: 2024-12-31 13:46:59.154930

value_counts of unit:
celsius 841988
bar 421069
rpm 421056
mm/s 421047
kw 420957

value_counts of machine_id:
1 351670
5 340199
8 332906
2 314584
7 314095
3 295141
4 291938
6 285584

- Column Explanation -

timestamp: UTC timestamp of sensor reading. Readings taken every ~5 minutes during production. Enables time-series feature extraction.

machine_id: Which machine generated the reading. Each machine has characteristic sensor patterns (+/- offset).

sensor_id: Composite ID like "3_temp_zone1" (machine_sensor_type). Unique per sensor per machine. Allows tracking specific sensor health.

value: Numerical reading. **30% are NULL** (realistic sensor dropout). Contains outliers and drift patterns. Examples:

- temp_zone1/2: 180-220°C (process temp, higher = defect risk)
- pressure_main: 5-8 bar (hydraulic pressure)
- vibration: 0.5-2.5 mm/s (bearing health indicator)
- speed: 800-1200 rpm (should match product spec)
- power_consumption: 15-35 kW (efficiency indicator)

unit: Measurement unit (celsius, bar, mm/s, rpm, kw). Required for proper interpretation.

2. Production Log Table — Column Overview

production table shape: (8000, 10)

#	Column	Dtype	Null-count	Null %	Unique value	Range	Percentile
1	production_id	object	0	0	8000	-	-
2	start_time	object	0	0	7759	-	-
3	end_time	object	0	0	7947	-	-
4	product_id	object	0	0	12	-	-
5	machine_id	int64	0	0	8	1 - 8	2 : 4: 8
6	operator_id	object	0	0	15	-	-
7	batch_size	int64	0	0	102	(-1) - 103	39 : 49 : 59
8	shift	object	0	0	3	-	-
9	environment_temp	float64	0	0	204	8:30 - 31.20	16.70 : 20.10 : 23.20
10	Environment_humidity	float64	0	0	456	22.80 - 76.50	43.40 : 49.90 : 56.70

Range of start_time:

Start: 2024-01-01 06:09:00

End: 2024-12-30 23:53:00

Range of end_time:

Start: 2024-01-01 16:04:00

End: 2024-12-31 23:11:00

value counts of product_id:

PRD_006 717

PRD_004 691

PRD_007 688

PRD_010 672

PRD_001 672

PRD_011 671

PRD_008 668

PRD_002 668

PRD_003 668

PRD_005 640

PRD_012 628

PRD_009 617

value counts of shift:

afternoon 3189

morning 2818

night 1993

- Column Explanation -

production_id: UUID for each production run. Primary key linking sensors, quality, and context.

start_time & end_time: Production window timestamps. Duration depends on batch size and product complexity. Used for time-series analysis and shift patterns.

product_id: What was manufactured. Foreign key to product_catalog. Determines difficulty and expected quality.

machine_id: Which machine ran the job (1-8). Links to maintenance history and sensor readings. Each machine has different baseline reliability.

operator_id: Who ran the machine. Links to operator experience/skills. Human factor in quality.

batch_size: Number of units produced. Range: ~20-80. Larger batches take longer. Affects per-unit metrics.

shift: morning/afternoon/night. The night shift has a 5% higher defect rate (fatigue factor). Distribution: 35%/40%/25%.

environment_temp: Celsius during production. Range: 13-28°C with seasonal patterns. Temps outside 18-25°C increase failures by 8%.

environment_humidity: Relative humidity %. Range: 30-70% with seasonal inverse pattern to temp. Outside of 30-70% adds 6% failure risk.

3. Product Catalog Table — Column Overview _____

product table shape: (12, 5)

#	Column	Dtype	Null-count	Null %	Unique value	Range	Percentile
1	product_id	object	0	0	12	-	-
2	product_family	object	0	0	4	-	-
3	tolerance_spec_1	float64	0	0	4	0.03 - 0.06	0.03 : 0.04 : 0.05
4	tolerance_spec_2	float64	0	0	4	0.04 - 0.09	0.04 : 0.06 : 0.08
5	recommended_speed	int64	0	0	4	65 - 155	87.50 : 125 : 155

value count of product_family:

BasicWidget	5
ComplexAssembly	3
StandardPart	2
PrecisionGear	2

- Column Explanation -

product_id: Unique identifier (PRD_001, PRD_002...). Primary key for joining to production runs.

product_family: Category of product being manufactured. Four types:

- BasicWidget (40%) - simpler products, looser tolerances
- PrecisionGear (30%) - requires more precision
- ComplexAssembly (20%) - hardest to manufacture, tightest specs
- StandardPart (10%) - medium complexity

tolerance_spec_1 & tolerance_spec_2: Manufacturing tolerance requirements in millimeters. Range: 0.01-0.06. Tighter tolerances (lower values) are harder to achieve and correlate with higher defect rates.

recommended_speed: Optimal production speed in units/hour. Range: 50-200. Complex products run slower. Deviating from this speed increases failure risk.

4. Quality Inspection Table – Column Overview _____

quality table shape: (8000, 7)

#	Column	Dtype	Null-count	Null %	Unique value	Range	Percentile
1	product_id	object	0	0	8000	-	-
2	inspect_time	object	0	0	7940	-	-
3	inspector_id	object	0	0	5	-	-
4	PIN_HOLES	int64	0	0	2	0 - 1	0 : 0 : 1
5	num_defects	int64	0	0	4	0 - 3	0 : 0 : 1
6	defect_types	object	0	0	157	-	-
7	severity_score	int64	0	0	7	0.0 - 9.0	0 : 0 : 5

Range of inspect_time:

Start: 2024-01-01 16:20:00

End: 2024-12-31 23:26:00

value counts of PIN_HOLES:

0 5528

1 2472

value counts of num_defects:

0 5528

1 842

2 825

3 805

5 most value counts of defect_types:

[] 5528

["incomplete_weld"] 154

["discoloration"] 150

["dimensional_error"] 148

["contamination"] 135

– Column Explanation –

production_id: Links back to production run. One inspection per production run.

inspect_time: When inspection occurred, 5-30 minutes after production ended.

inspector_id: Who performed inspection (INS_01 through INS_05). Could reveal inspector bias patterns.

PIN_HOLES: PRIMARY TARGET VARIABLE. Binary 0/1 indicating pass/fail. 1 = defect detected, 0 = passed. ~15% defect rate (class imbalance). Has 7% label noise (mislabeling).

num_defects: Count of defects found (0-3). Only non-zero when PIN_HOLES=1. Indicates severity through quantity.

defect_types: JSON array of defect categories found: surface_crack, dimensional_error, contamination, incomplete_weld, burr, discoloration. Empty for passing units. Enables root cause analysis.

severity_score: 0-10 scale. 0 for passing units, 4-10 for failures. Could be used for regression instead of classification, or for weighted metrics.

5. Maintenance Log Table — Column Overview

maintenance table shape: (83, 6)

#	Column	Dtype	Null-count	Null %	Unique value	Range	Percentile
1	maintenance_id	object	0	0	83	-	-
2	machine_id	int64	0	0	8	1 - 8	3 : 5 : 7
3	maintenance_time	object	0	0	82	-	-
4	type	object	0	0	2	-	-
5	downtime_minutes	int64	0	0	68	52 - 309	105 : 126 : 145.50
6	parts_replaced	object	0	0	10	-	-

Range of maintenance_time:

Start: 2024-01-01 01:00:00
End: 2024-12-29 00:00:00

value counts of type:

preventive 74
corrective 9

top 5 value counts of parts_replaced:

[] 40
["filter", "lubricant"] 34
["motor"] 2
["motor", "bearing"] 1
["sensor"] 1

– Column Explanation –

maintenance_id: Unique maintenance event ID (MNT_00001...).

machine_id: Which machine was serviced (1-8). Links to production runs and sensors.

maintenance_time: Timestamp when maintenance occurred. Used to calculate "days since last maintenance" - a key failure predictor.

type:

- preventive: Scheduled maintenance every 30-45 days, ~120 min downtime
- corrective: Unscheduled repairs after breakdowns, ~240 min downtime, indicates reliability issues

downtime_minutes: Production stopped during maintenance. Affects capacity planning. Corrective maintenance causes longer downtime.

parts_replaced: JSON array of components replaced (bearing, sensor, valve, motor, belt, filter, lubricant). Empty array means service only. Indicates wear patterns.

6. Operators Details Table — Column Overview

operators table shape: (15, 5)

#	Column	Dtype	Null-count	Null %	Unique value	Range	Percentile
1	operator_id	object	0	0	15	-	-
2	name_hash	object	0	0	15	-	-
3	experience_months	int64	0	0	4	6 - 120	15 : 36 : 36
4	certifications	object	0	0	6	-	-
5	training_date	object	0	0	4	-	-

Range of training_date:

Start: 2014-02-22

End: 2023-07-05

value counts of certifications:

["CERT_0", "CERT_1"]	4
["CERT_0", "CERT_1", "CERT_2"]	3
[]	2
["CERT_0", "CERT_1", "CERT_2", "CERT_3"]	2
["CERT_0", "CERT_1", "CERT_2", "CERT_3", "CERT_4"]	2
["CERT_0"]	2

- Column Explanation -

operator_id: Unique operator identifier (OP_001, OP_002...). Links to production runs.

name_hash: Anonymized operator name as hex hash. Represents PII protection in real systems.

experience_months: Time working in manufacturing. Range: 6-120 months. Higher experience → lower defect rates (skill effect).

certifications: JSON array of certifications (e.g., ["CERT_0", "CERT_1"]). More certifications generally align with experience. Affects quality through training.

training_date: Date of most recent training. Calculated backward from experience duration. Shows last skill update.